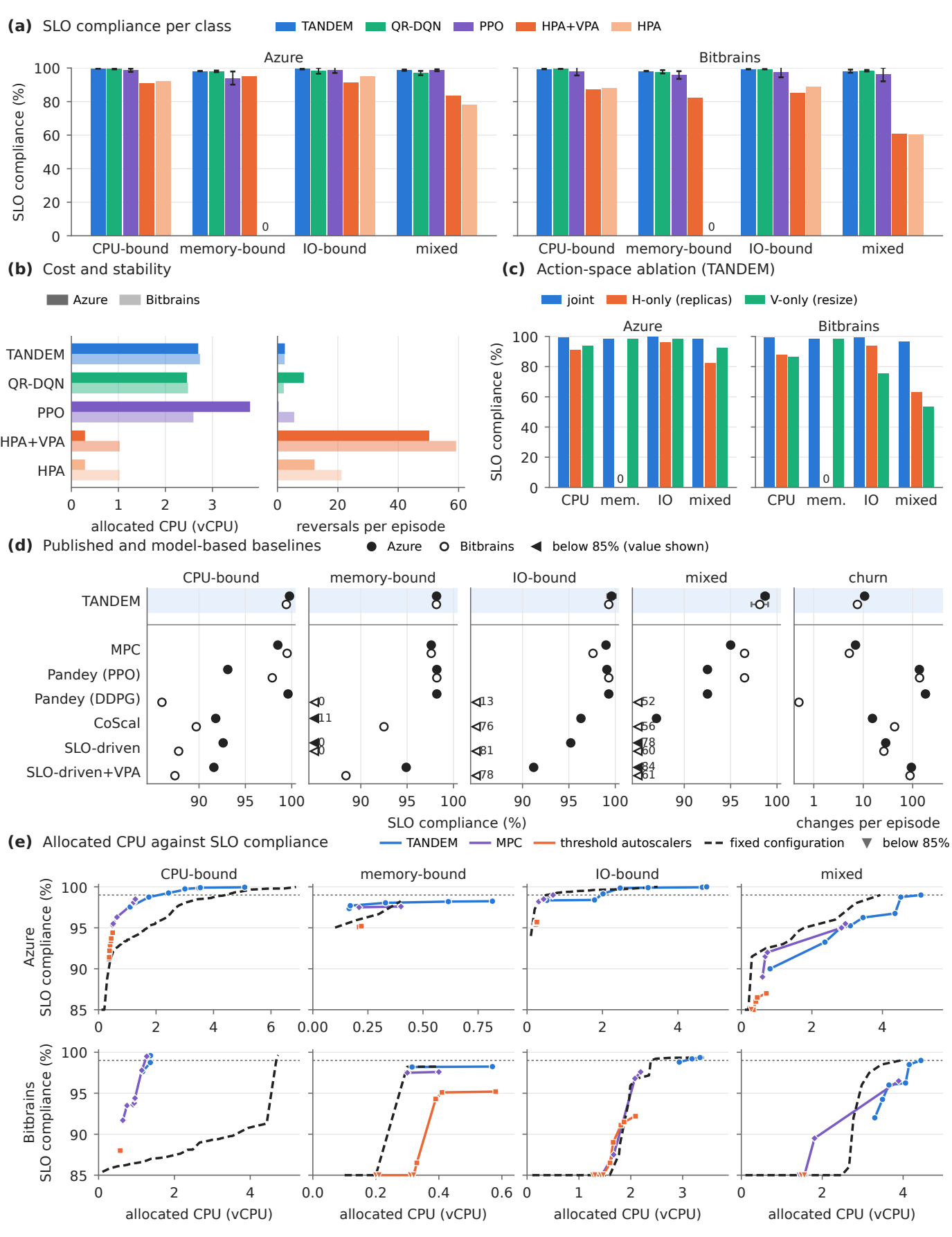
(a) SLO compliance per class
TANDEM
QR-DQN
PPO
HPA+VPA
HPA
Azure
Bitbrains
SLO compliance (%)
CPU-bound
memory-bound
IO-bound
mixed
(b) Cost and stability
allocated CPU (vCPU)
reversals per episode
(c) Action-space ablation (TANDEM)
joint
H-only (replicas)
V-only (resize)
CPU
mem.
IO
(d) Published and model-based baselines
below 85% (value shown)
churn
MPC
Pandey (PPO)
Pandey (DDPG)
CoScal
SLO-driven
SLO-driven+VPA
changes per episode
(e) Allocated CPU against SLO compliance
threshold autoscalers
fixed configuration